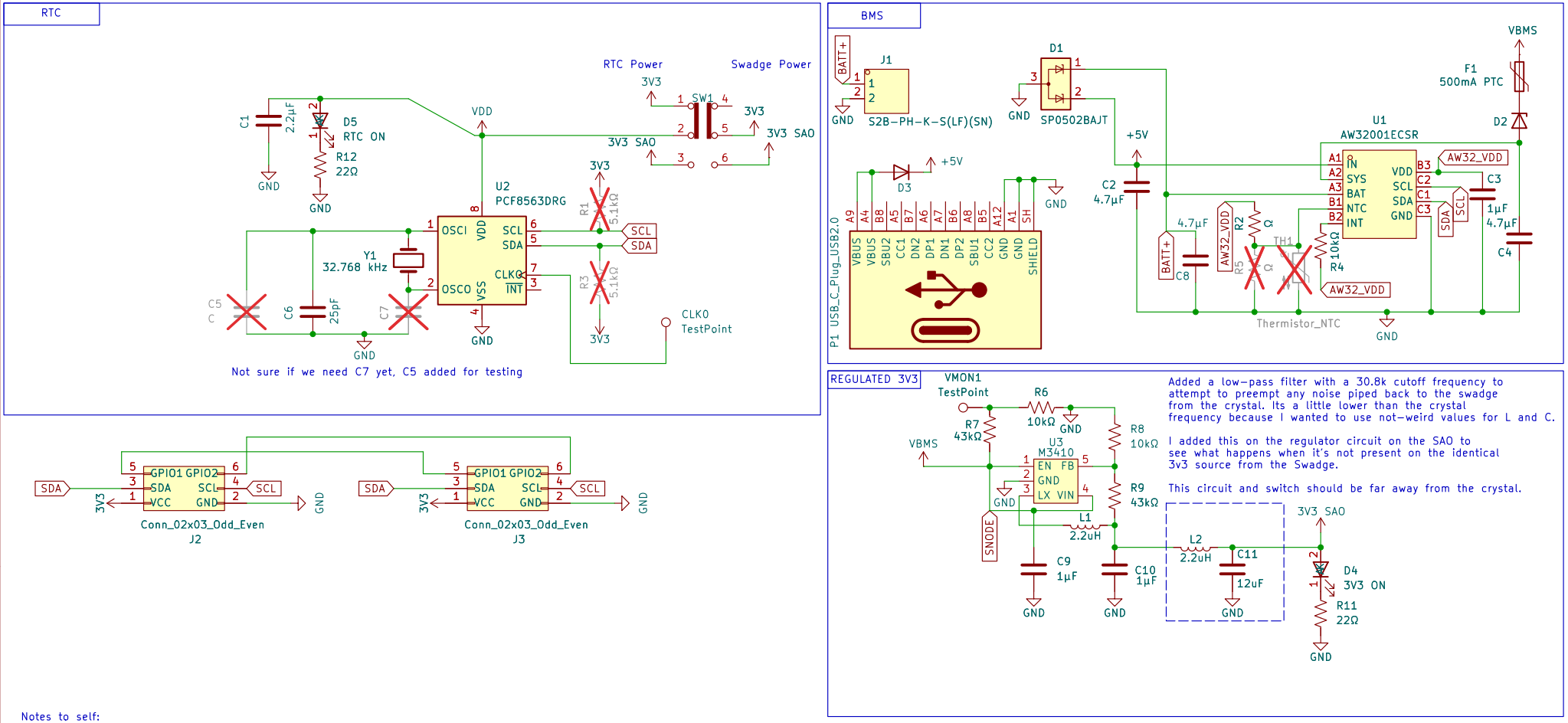


Notes to self:
1. After power-on, the initial accuracy can be adjusted by testing the output frequency of CLKOUT and adjusting the matching capacitor value of the OSCI pin.
If CLKOUT > 32.768kHz, the OSCI capacitor value needs to be increased; if CLKOUT < 32.768kHz, the capacitor value needs to be decreased.

2. Generally, if using a 32.768kHz crystal oscillator with a load capacitor of 12.5pF, approximately a 25pF matching capacitor needs to be connected to the OSCI pin to adjust the initial accuracy. Also added C7 on OSCO. Final values TBD

3. If noise or interference signals occur on the VDD power line, it is recommended by the RTC chip mfr to add resistors and capacitors to the power line for filtering.
Currently C1 is the filter cap for the Swadge 3v3.

4. Crystal requirements from RTC datasheet: f = 32.768 kHz, CL = 12.5 pF, ESR = 50-100k

5. I added a 500mA PTC fuse to test some short circuit protections.

6. I2C addresses:
PCF8563 = 0x51
AW32001 = 0x49
Set NTC to DISABLE; this is not the default setting for this chip. TH1 = DNP, R1 = 0, R2 = DNP.

7. This will back-power the Swadge. Use the switch to choose between SAO regulated power or Swadge regulated power. Please be careful and remove the power source from the Swadge before switching over to SAO power.

8. USB is for charging only.

9. Datasheet: https://www.nxp.com/docs/en/data-sheet/PCF8563.pdf

Requirements grow
like ivy on a garden wall.
The plan disappears.

Sheet: /
File: sao_rtc.kicad_sch

Title: SAO for RTC evaluation

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